Course Name: Blockchain and its Applications (NOC26_CS34)

Assignment 5 - Week 5 (Feb 2026)

TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1

Which of the following factors generally limit the scalability of permissionless blockchain consensus mechanisms?

a. Requirement for a global agreement among nodes
b. Centralized validator selection
c. High communication and computation overhead
d. Absence of cryptographic security

Answer: (a) and (c)

Detailed solution:
Permissionless blockchains require participation and verification by a large number of nodes. Achieving global consensus introduces significant communication and computation overhead, which limits scalability. Validator selection is decentralized, and cryptographic security is a core feature, not a limitation.

QUESTION 2

Which consensus mechanism relies on trusted hardware environments to ensure fairness in block proposer selection?

a. Proof of Stake
b. Proof of Burn
c. Proof of Elapsed Time
d. Practical Byzantine Fault Tolerance

Answer: (c)

Detailed solution:
Proof of Elapsed Time (PoET) uses trusted execution environments such as Intel SGX to generate verifiable random wait times, ensuring fair leader selection without intensive computation.
Please refer to Week 5, Lecture 21.

QUESTION 3

Which of the following properties is a key advantage of stake-based (PoS) consensus mechanisms compared to computation-based (PoW) mechanisms?

a. Complete elimination of forks
b. Reduced energy consumption
c. Guaranteed transaction finality
d. Removal of the need for digital signatures

Answer: (b)

Detailed solution:
Stake-based consensus mechanisms, such as Proof of Stake, do not rely on energy-intensive cryptographic puzzles. As a result, they significantly reduce energy consumption compared to computation-based mechanisms like Proof of Work.

QUESTION 4

In a PoW-based blockchain system like Bitcoin, when two valid blocks are mined at the same time, which rule does the network primarily use to resolve the fork and identify the authoritative chain?

a. Number of transactions in each block
b. Total accumulated computational work (heaviest chain)
c. Size of the blockchain ledger
d. Time since genesis block creation

Answer: (b)

Detailed solution:
The longest-chain rule selects the chain with the highest cumulative computational work. This ensures that the chain requiring the most effort to produce is treated as authoritative.

QUESTION 5

Which component of the Ethereum architecture ensures deterministic execution of smart contracts across all nodes?

a. Ethereum Client
b. Ethereum Virtual Machine
c. Ethereum Wallet
d. Ethereum Miner

Answer: (b)

Detailed solution:
The Ethereum Virtual Machine (EVM) executes smart contract bytecode in a deterministic manner, ensuring that all participating nodes reach the same result when executing the same contract.

QUESTION 6

Which of the following operations on Ethereum requires gas payment and inclusion in a block?

a. Querying an account balance
b. Reading a public contract variable
c. Executing a state-modifying function
d. Simulating contract execution locally

Answer: (c)

Detailed solution:
State-modifying operations change the blockchain state and therefore require a transaction to be sent to the network, consume gas, and be included in a block.

QUESTION 7

Which Ethereum transaction parameter ensures transaction ordering and prevents replay attacks for a given account?

a. Gas
b. Value
c. Nonce
d. Data

Answer: (c)

Detailed solution:
The nonce uniquely identifies each transaction sent from an account and enforces sequential ordering, preventing replay of old transactions.

QUESTION 8

Which JSON-RPC method is used to retrieve the Ether balance of an Ethereum account?

a. eth_call
b. eth_getBalance
c. eth_getTransactionByHash
d. eth_sendTransaction

Answer: (b)

Detailed solution:
The eth_getBalance JSON-RPC method returns the balance of a specified Ethereum address at a given block.

QUESTION 9

Which consensus mechanism requires participants to irreversibly destroy cryptocurrency to demonstrate long-term commitment to the network?

a. Proof of Work (PoW)
b. Proof of Burn (PoB)
c. Proof of Stake (PoS)
d. Proof of Elapsed Time (PoET)

Answer: (b)

Detailed solution:
Proof of Burn requires participants to send cryptocurrency to a verifiably unspendable address, effectively destroying the coins to prove commitment.

QUESTION 10

Which of the following Ethereum unit relationships is CORRECT?

a. 1 Ether = 10^9 Wei
b. 1 Gwei = 10^9 Wei
c. 1 Ether = 10^6 Gwei
d. 1 Wei = 10^{-9} Ether

Answer: (b)

Detailed solution:
Gwei is a commonly used denomination in Ethereum, defined as 10^9 Wei. Gas prices are typically specified in Gwei.